Potential Impact Assesment

Minor AI for Society

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1. Document History

Table 1. Document History

Version	Date	Status	Author	Description	
1.0	09-03-2025	Initial Version	Casey Stijlaart	Initial version	
1.1	12-03-2025	First Version	Casey Stijlaart	Added mitigation strategies and sustainability considerations	
2.0	12-03-2026	Completed First Version	Casey Stijlaart	Restructured to reflect Responsible AI guidelines, future scenarios, and design reflections	

2. Purpose & Context

2.1. Impact on Society

The AI-powered Smart Desk Plant Monitoring system aims to reduce the “pain” of plant neglect and unsuccessful plant care, a common challenge for individuals in home and office environments By providing timely and personalized recommendations, the system addresses the problem of plant health deterioration due to inconsistent environmental awareness.

I am confident that the system addresses the right problem because the data collected (e.g., soil moisture, light, temperature, and humidity) directly informs actionable care recommendations. The system’s design focuses on user empowerment and education, rather than automation, to ensure that users remain engaged and informed about their plants’ needs. The intended societal outcome is improved environmental literacy, reduced plant loss, and increased engagement with small-scale environmental monitoring. I aim to avoid outcomes where users over-rely on AI, leading to reduced autonomy or misunderstanding of natural plant care processes.

Improvements considered: Implement adaptive recommendation messaging that emphasizes user judgment, integrate reminders to check sensor readings manually, and provide educational content about plant physiology.

2.2. Domain Understanding

Currently, plant care relies heavily on personal knowledge, habit, or generic online advice. Users vary widely in expertise, environmental constraints, and attention levels. Successful outcomes require consistent monitoring and correct interpretation of environmental signals; poor outcomes include plant death or chronic stress. Existing solutions (apps, watering reminders, environmental sensors) often fail due to over-reliance on cloud services, lack of context-specific guidance, or poor usability.

Assumptions: Users are willing to interact with a local dashboard, interpret suggestions responsibly, and accept minimal hardware installation. These will be validated through usability testing, and iterative prototypes.

Professional standards and constraints: Safety in electrical components, proper data protection, ethical AI communication, and sustainability considerations. Major uncertainties include user adherence, sensor accuracy, and long-term maintenance.

Improvements: Design for modular sensor replacement, ensure offline operation for privacy, and include clear instructions for user oversight.

2.3. Stakeholders

Primary stakeholders: Individuals who own plants at home or workspaces, affected by improved plant care and educational value.

Secondary stakeholders: Hobby gardeners, educators, and tech enthusiasts, affected by exposure to embedded AI applications and environmental awareness.

Indirect stakeholders: Environmental organizations, institutions exploring indoor environmental monitoring, or bystanders benefiting from improved ecological literacy.

Planned consultation: Prototype testing, and feedback loops to incorporate stakeholder insights.

Improvements: Broader engagement with diverse user groups to ensure recommendations are clear and actionable for novices and experts alike.

3. Societal Impact

3.1. Privacy

The system processes environmental data locally, no personal identifiers are collected. In case of dashboard integration requiring external access, anonymization and encryption are enforced.

Potential risks: Indirect profiling if users' patterns are combined with other datasets.

Improvements: Introduce strict data retention policies, anonymization safeguards, and user control over any transmitted data.

3.2. Data

Sensor data is objective but limited in context. Missing context or calibration issues may bias recommendations.

Mitigations: Continuous calibration, validation of sensors, and iterative model updates. Insights remain sustainable via adaptive models and regular data review.

Trust and consent: Users are informed about data use, consent is explicit for any external communication.

Improvements: Regularly review and document model assumptions, provide transparency in algorithm limitations.

3.3. Transparency

The system communicates plant recommendations as suggestions, not authoritative commands. Limitations and operational principles are clearly documented in the dashboard.

Improvements: Add FAQs, visual explanations of AI reasoning, and feedback channels for users to challenge outputs.

3.4. Inclusivity

Access may be limited by cost, technical literacy, or language barriers. Bias is minimal, as the system is largely universal for plant types, but cultural context may influence interpretation.

Improvements: Multi-language support, accessible interface design, and open-source options for wider adoption.

3.5. Sustainability

Local TinyML operation reduces cloud energy consumption, but materials like sensors and microcontrollers have environmental cost.

Improvements: Select durable, low-power components, design for repairability and modularity, and encourage recycling.

3.6. Hateful and Criminal Actors

The technology is low-risk for misuse, potential abuse includes misrepresentation of sensor data in shared environments.

Mitigations: Security of local devices, firmware validation, and clear disclaimers.

3.7. Human Values

Strengthened values include environmental awareness, autonomy in decision-making, and learning about AI and plants.

Potential compromises includes over-reliance on recommendations could reduce critical engagement, or stress if users feel pressured to follow AI advice.

Improvements: Emphasize educational framing, provide autonomy in decision-making, and avoid gamification that pressures compliance.

4. Future Scenarios

4.1. Utopian Future

20–50 years ahead, widespread adoption of embedded AI plant monitoring contributes to urban greenery, reduced energy in agriculture, and greater environmental literacy. Users integrate sustainable practices and autonomous AI assists without reducing personal agency. Governance and community sharing maintain transparency and equitable access.

4.2. Dystopian Future

AI-driven monitoring is over-adopted, leading to dependency on devices, loss of traditional plant knowledge, and electronic waste proliferation. Corporate control or proprietary lock-in could limit user agency and accessibility, concentrating benefits to affluent groups. Misuse of aggregated environmental data could enable indirect profiling of households.

5. Impact Reflection

5.1. Transformative Perspectives

TBD: Reflect on how the project has influenced your understanding of AI's role in everyday life, the importance of responsible design, and the potential for small-scale AI applications to contribute to broader societal goals.

5.2. In Practice

TBD: Reflect on how the project evolved, what was learned about the problem space, and how the design adapted to address unforeseen challenges.